

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

**STUDENT
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Applications Paper and Planning Question
Paper 3

9648/03

MONDAY 02 SEPTEMBER 2013
2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
1 [15]	
2 [10]	
3 [15]	
Planning	
4 [12]	
Free-response	
5a [7]	
5b [6]	
5c [7]	
TOTAL	72

Answer **all** questions.

- 1 (a) Explain what is meant by a restriction enzyme.

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..... [2]

- (b) Fig. 1.1 illustrates part of a DNA sequence that is recognised by a restriction enzyme. Complete the sequence by filling up the blanks in Fig. 1.1. [1]

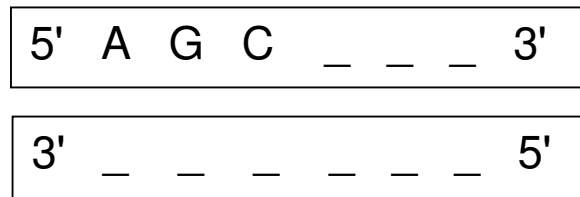


Fig. 1.1

- (c) What do the 5' and 3' ends in Fig. 1.1 represent?

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..... [2]

Fig. 1.2 illustrates a typical plasmid used in genetic engineering.

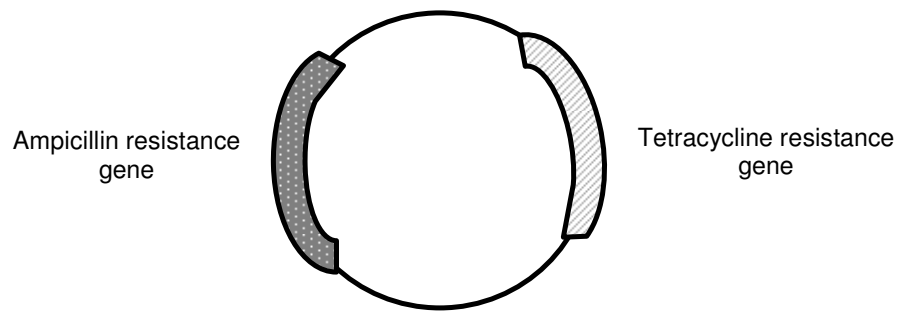


Fig. 1.2

(d) The plasmid shown in Fig. 1.2 lacks two features that are essential for genetic engineering. Add these two features to the plasmid in Fig. 1.2 and label them. [2]

(e) Explain the significance of one of the features added in part (d).

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..... [1]

A dot blot is a simplified version of Southern Blotting. It can be employed to test for the presence of gene mutations. In this molecular technique, undigested DNA is applied directly onto a membrane through a vacuum. These dots containing DNA will be hybridised to a sequence-specific oligonucleotide probe and be subjected to autoradiography. Dots which successfully hybridise to a probe will be seen as a dark dot on the autoradiogram.

- (f) Describe two key differences between the process of obtaining a dot blot and a Southern blot.

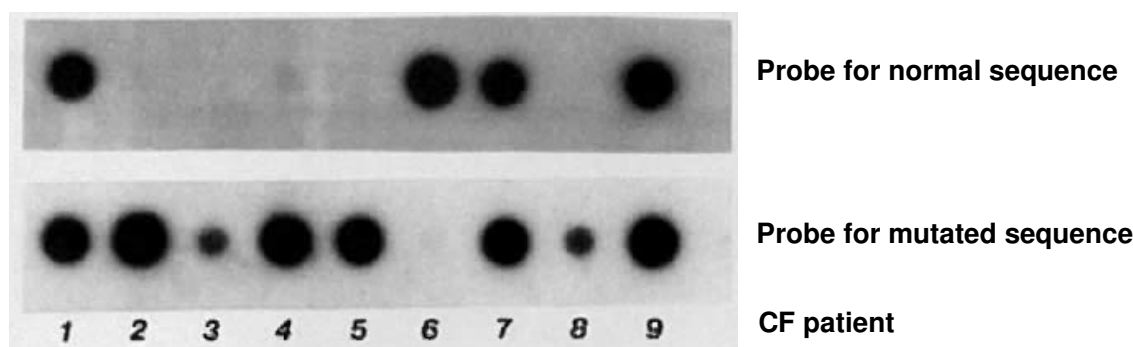
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..... [2]

Cystic fibrosis (CF) has been linked to several mutations. Two probes have been synthesized to hybridise with a mutated sequence associated with CF as well as the corresponding normal sequence. The dot blots of nine patients clinically diagnosed with CF are shown in Fig. 1.3.



Adapted from: <http://www.glowm.com/resources/glowm/cd/pages/v5/ch076/framesets/006f.html>

Fig. 1.3 Dot plot of 9 CF patients.

- (g) State one mutation linked to cystic fibrosis.

..... [1]

- (i) Account for the presence of two dots in individuals 1, 7 and 9.

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..... [2]

- (j) With reference to Fig. 1.3, explain why the mutation associated with CF is most likely to be recessive.

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..... [2]

[Total: 15]

- 2 A patient was found to have a defective gene that results in the failure to produce an important chemical substance, chemical substance X.

Three experiments were carried out to compare three different techniques for introducing chemical substance X into such patients:

Experiment 1: The chemical substance X was injected directly into the patient.

Experiment 2: The gene coding for chemical substance X was packaged into a viral vector that was modified to render it harmless. The viral vector containing the gene was injected into the patient.

Experiment 3: The gene coding for chemical substance X was placed into a protein capsule, which was then inserted into the tissues of the patient.

Fig. 2.1 shows the results from these experiments.

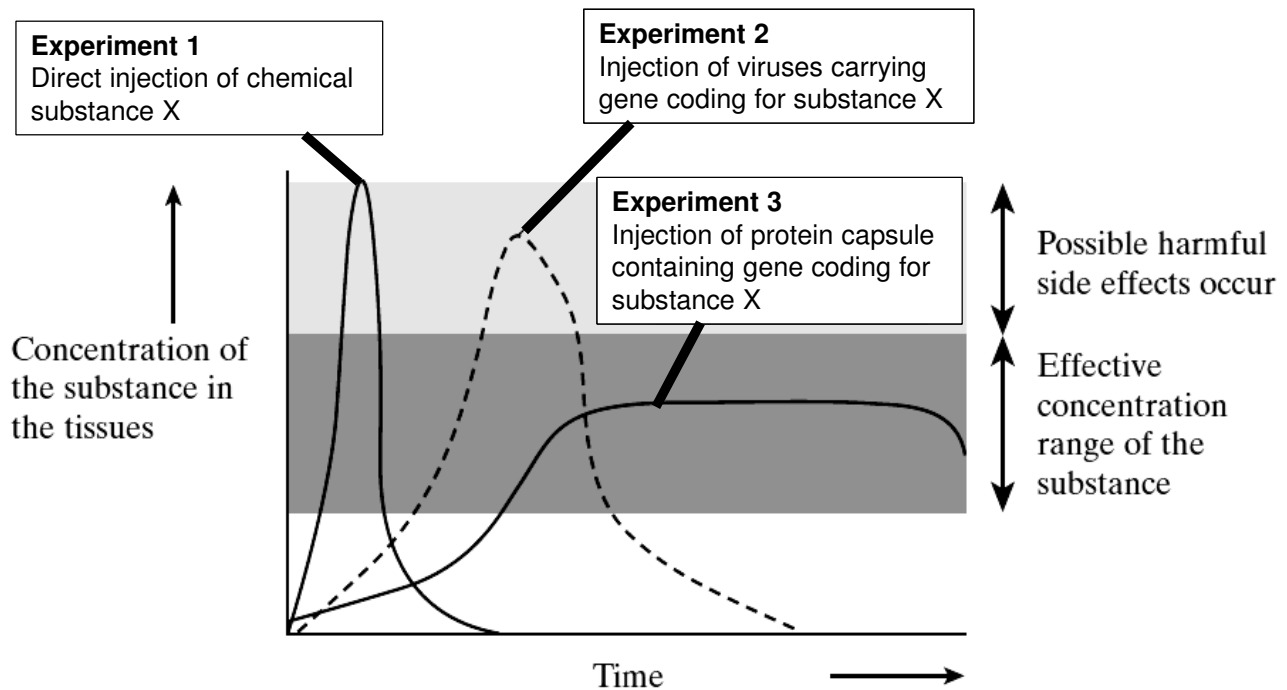


Fig. 2.1

- (a) (i) Describe the results of the three experiments.

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- (ii) Using the information in the graph, suggest two advantages and one disadvantage of the protein capsule method, as compared to the other methods.

Advantages:

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..... [2]

Disadvantage:

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Duchenne muscular dystrophy (DMD) is a common lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres.

Two possible treatments have been proposed to correct the deficiency of dystrophin:

- (1) Transplant normal cells from genetically compatible donors into the patient;
- (2) Gene therapy.

Gene therapy involves inserting the gene coding for dystrophin into an adenovirus and injecting the recombinant adenovirus into the patient's muscle.

- (b) Gene therapy using the recombinant adenoviruses did not provide a permanent cure for DMD. The trials showed that the transgene expression declined after about 2 weeks and was negligible after only 4 weeks. Explain this phenomenon.

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- (c) Suggest a suitable vector for delivery of genes designed to kill cancer cells. Explain your choice.

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..... [2]

[Total: 10]

3 (a) Genetic engineering has been used to improve the quality and yield of many crop plants. One method of genetic engineering involves modifying Ti plasmid from a bacterium, *Agrobacterium tumefaciens*, and transferring the modified Ti plasmid containing the transgene into crop plants. This method has been used for the production of *Bt* corn.

(i) State one other method that can be used for the genetic engineering of crop plants.

..... [1]

(ii) The Ti plasmid is usually modified to contain an antibiotic resistance gene (e.g. ampicillin resistance gene).

Explain the purpose of this modification.

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The successfully genetically engineered crop plant cells may be stimulated to grow into plantlets using tissue culture.

(iii) Describe how a plantlet may be produced from cells in tissue culture.

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..... [2]

(iv) Describe two advantages of using tissue culture to clone genetically modified plants.

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..... [2]

- (b) A number of different crop plants have been genetically engineered to express a gene for an insecticidal toxin (*Bt* toxin) from a bacterium, *Bacillus thuringiensis*.

The *Bt* toxin binds to cell membranes of the cells in the insect gut. The receptor for the toxin has been identified as a membrane-anchored protein encoded by the cadherin gene. *Bt* toxin forms ion channels that cause an efflux of potassium ions from the insect gut cells, leading to cell lysis.

The cotton bollworm, *Helicoverpa armigera*, is one of the most serious insect pests of cotton in Asia, Australia, and Africa. A major deletion of the cadherin gene was associated with a high level of *Bt* toxin resistance in *H. armigera*. Three alleles (*r1*, *r2*, and *r3*) of the cadherin gene in the resistant strain have been identified:

- *r1* allele lacks 24-bp
- *r2* allele has a 202-bp deletion
- *r3* allele has a 126-bp deletion

- (i) Explain how deletions of the cadherin gene could confer resistance to *Bt* toxin in *H. armigera*.

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A survival test was conducted on larvae of different genotypes to assess their resistance to *Bt* toxin.

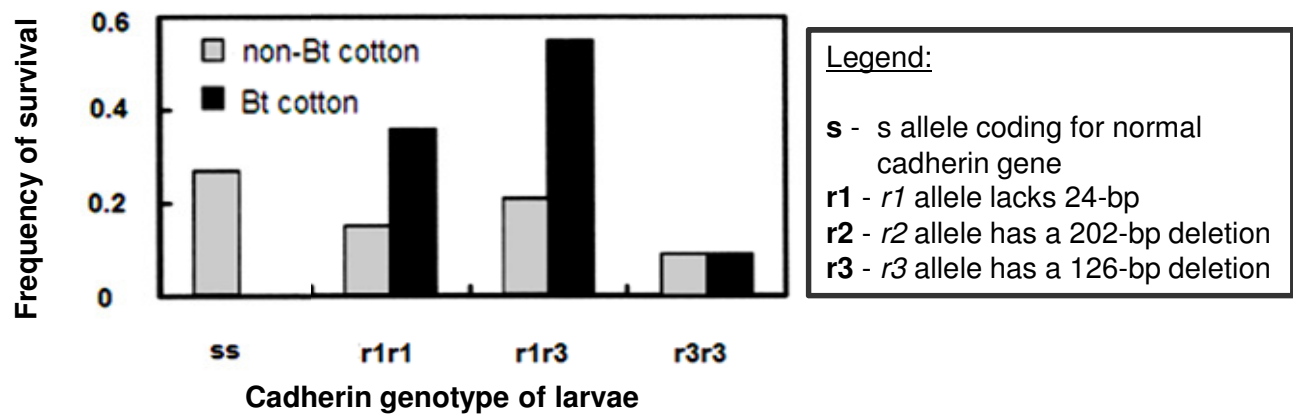


Fig. 3.1

(ii) Suggest why larvae of genotype ss was included in this test.

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 [1]

(iii) Describe the effects of the genotypes r1r1, r1r3 and r3r3 on the resistance of larvae to Bt toxin.

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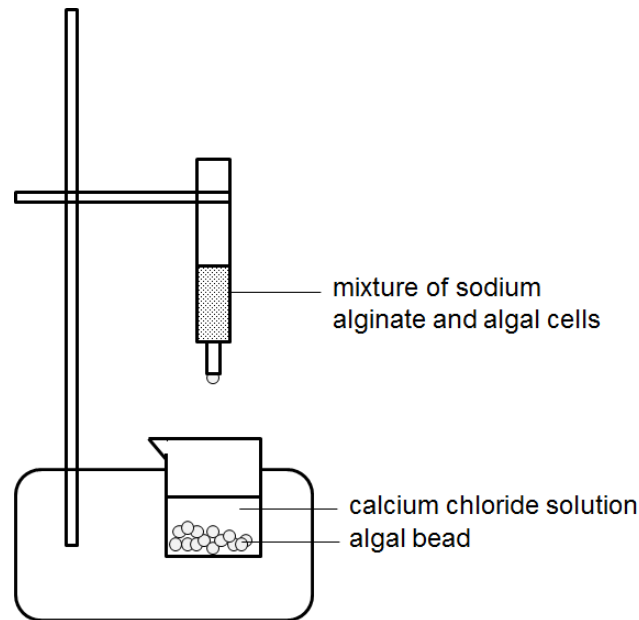
 [3]

[Total: 15]

4 Planning question

You are required to investigate the rate of photosynthesis on a certain species of algae, *Scenedesmus Chlorella*.

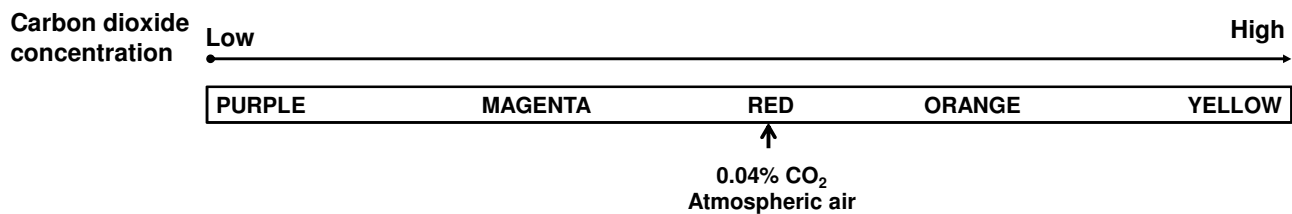
Algae are tiny and difficult to work with directly in water. Hence, sodium alginate is usually used to trap large numbers of algal cells into agar beads so that the cells are all contained in one piece.



Source: Copyright Science & Plants for Schools: www.saps.org.uk

Fig. 4.1

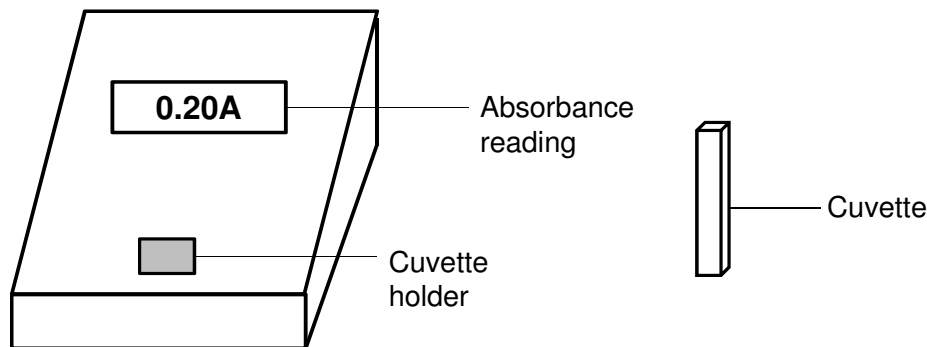
Hydrogen carbonate indicator can be used to measure the rate of photosynthesis as it is very sensitive to changes in carbon dioxide level. The indicator is red when the amount of CO_2 in the solution is in equilibrium with atmospheric air, becomes orange to yellow with increased CO_2 in the solution and changes from red through magenta to deep purple as CO_2 is removed from the solution. This is represented in the diagram below.



Source: Copyright Science & Plants for Schools: www.saps.org.uk

Fig. 4.2

The colour changes in hydrogen carbonate indicator can be measured objectively using a colorimeter. A colorimeter measures the absorbance of the solution in the cuvette. The cuvette holds the working solution.



Source: Copyright Science & Plants for Schools: www.saps.org.uk

Fig. 4.2

Different colours absorb different wavelengths of light to a different extent. The darker the colour, the greater the absorbance value (A) will be. The known absorbance values (A) of the following colours of the hydrogen carbonate indicator are shown in the table below.

Table 4: Table of known absorbance values

Color of Hydrogen Carbonate Indicator	Absorbance of Colorimeter/ A
Yellow	0.10
Orange	0.20
Red	0.30
Magenta	0.50
Purple	0.70

A = Absorbance units

Using all the information above, you are required to plan, but not carry out, an investigation into:

The effect of light intensity on the rate of photosynthesis in the algae, *Scenedesmus Chlorella*

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- 150 *Scenedesmus Chlorella* algal beads
- 100 cm³ of hydrogen carbonate indicator solution containing 387 parts per million (ppm) CO₂, which is at equilibrium with the 0.04% atmospheric CO₂.
- 7 cuvettes
- Colorimeter with a digital display to 2 decimal places and a cuvette holder.
- 7 glass tubes with screw-on-lids. The walls of these tubes are transparent.
- A 150 W bench lamp which emits very little heat.
- 5 cm³ syringe
- Stop watch.
- Ruler.

Your plan should:

- have a clear and helpful structure such that the method proposed can be repeated by anyone reading it,
- be illustrated by relevant diagram(s) to show, for example, the arrangement of the apparatus used,
- include layout of results tables and graphs with clear headings and labels,
- include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.

[Total: 12]

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[illegible]

[illegible]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a) Describe the unique features of stem cells and explain the normal functions of stem cells in a living organism. [7]
- (b) Explain what is meant by restriction fragment length polymorphism (RFLP) and how it can be detected. [6]
- (c) Using a **named example**, explain how RFLP analysis can be used for disease detection. [7]

[Total: 20]

END OF PAPER

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